

KIDS SCIENCE PODCAST AI PRODUCTION SPECIFICATION

Full-Stack SOP & Skill Guideline: Qwen3-TTS Engine + Audio Director Workflow

Format: Solo AI Monologue

TTS Engine: Qwen3-TTS (1.7B)

Target: Kids Aged 7–10

Theme: What Happened On This Day

1. EXECUTIVE SUMMARY & AUTOMATED WORKFLOW

This specification outlines the complete production framework for a daily children's podcast utilizing a **100% locally hosted AI Voice Host (Qwen3-TTS)**. The Producer acts as the AI Sound Director and Quality Gatekeeper, turning structured markdown scripts into fully mixed, high-engagement audio episodes.

THE 4-STEP AUTOMATED PIPELINE:

- Scripting (SKILL.md):** Modular 10–12 min script generation with explicit audio tags.
- Lexicon Tagging:** Pinyin/Zhuyin preprocessing for scientific terms and homophones.
- Local Qwen3-TTS Inference:** Synthesize audio via Qwen3-TTS 1.7B CustomVoice with locked seed & prompt.
- Sound Design & Mastering:** Layer SFX/BGM acoustic masking, normalize to -16 LUFS, and push to RSS.

2. HOST PERSONA & AI VOICE ENGINE SETUP

Target Audience & Persona

- Demographic:** 7–10-year-olds (Elementary Schoolers) seeking fun, logical explanations of real-world science.
- Host Persona:** "The Cool Science Explorer" — an energetic, time-traveling science detective speaking as a peer.
- Strict Ban List:** Zero baby-talk ("doggy", "tummy-wummy"), no monotone academic lecturing, no flat robotic delivery.

Local AI Engine Architecture (Qwen3-TTS)

- Primary Model:** Qwen3-TTS-12Hz-1.7B-CustomVoice (4–8 GB VRAM) for maximum prosody, emotion, and life-like inflection.
- Fallback / Edge Model:** Qwen3-TTS 0.6B or GGUF quantization for low-resource CPU execution.
- Voice Profile Locking:** Lock the reference audio (3–5s clean voice prompt) and inference seed parameter to guarantee 100% IP voice consistency.
- Audio In-filling:** Use Qwen3-TTS localized re-synthesis to repair single mispronounced lines without regenerating full episodes.

3. AUDIO-FIRST WRITING & TAGGING STANDARDS

Scriptwriters and AI generators must format scripts with standardized audio direction tags:

- [SFX: <Description>]** — Sound effects for scene building & acoustic masking (e.g., **[SFX: Time machine zoom]**).
- [BGM: <Mood/Action>]** — Music background cues (e.g., **[BGM: Playful investigative synth]**).
- [Voice: <Tone/Speed>]** — Prosody instructions (e.g., **[Voice: Whisper in secret]** , **[Voice: High excitement]**).
- [Pause: <Duration>]** — Natural breathing/pacing control (e.g., **[Pause 2s]**).

4. UNIVERSAL 10–12 MINUTE EPISODE STRUCTURE

TIME BLOCK	MODULE NAME	SCRIPTING STRATEGY & AI PROSODY DIRECTIONS
0:00 – 1:30	1. Time Machine Hook	Start with a counter-intuitive mystery. Announce today's date and launch time machine. Hook in 15 seconds. <i>[Voice: Energetic, fast]</i>
1:30 – 5:00	2. Narrative Drama	Tell historical story (hero vs obstacle). Personify objects (bacteria, space probes). Build tension. <i>[Voice: Suspenseful]</i>
5:00 – 8:30	3. Scientific Deep-Dive	Core science in 2 steps . Mandatory: 1 everyday metaphor + 1 physical action prompt. <i>[Voice: Clear, steady]</i>
8:30 – 10:30	4. Modern World Twist	Reframe: How this impacts kids today. Bust a myth or reveal fun fact. <i>[Voice: Mind-blown, amazed]</i>
10:30 – 12:00	5. Outro & Mission	30s summary. Assign offline "Daily Science Observation Task". Closing catchphrase. <i>[Voice: Warm, encouraging]</i>

5. ACOUSTIC MASKING & SOUND DESIGN RULES

- **Acoustic Masking:** Place sound effects (**SFX**) over AI voice transition points to conceal synthetic tail-offs or unnatural artifacts.
- **EQ & Warmth Shaping:** Boost mid-bass (150–300 Hz) for voice warmth; filter frequencies above 16 kHz to remove digital harshness.
- **Loudness Standard:** Normalize final multitrack mix to **-16 LUFS** with -1.0 dB true peak limit.

6. QUALITY ASSURANCE CHECKLIST FOR AI SOUND PRODUCER

- ☐ **15-Second Hook:** Does the opening grab attention immediately with a mystery?
- ☐ **Lexicon Preprocessing:** Are proper nouns and homophones phonetic-tagged (Pinyin/Zhuyin)?
- ☐ **Voice IP Consistency:** Was the episode synthesized using locked Qwen3-TTS Voice Prompt & Seed?
- ☐ **Masking & Audio Density:** Are sound effects (**SFX**) embedded every 45–60 seconds?
- ☐ **Relatable Metaphor:** Is there at least ONE vivid everyday metaphor explaining the science?
- ☐ **Loudness Compliance:** Is the final multitrack export normalized to -16 LUFS?
- ☐ **Take-Home Task:** Is there a safe, offline physical observation task for the child?